Loan Default Analysis & Risk Assessment

Power BI Data Analytics Project Report

Made with GAMMA

Executive Summary

Objective

Analyze loan default patterns and financial risk factors using interactive Power BI dashboard

Focus

Identify key borrower characteristics contributing to default risk

Outcome

Data-driven framework for credit risk management and lending decisions

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Business Problem

Financial institutions face significant losses due to loan defaults. Identifying high-risk borrowers before approval is critical.

01

What factors contribute to loan defaults?

02

Which borrower profiles are most likely to default?

03

How do income, financial status, and demographics affect repayment?

04

Can borrowers be segmented into risk categories?

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Dataset Overview

Borrower Demographics

- Age, employment status
- Education level
- Marital and residential status

Financial Attributes

- Annual income, loan amount
- Credit score
- Debt-to-income ratio
- Existing liabilities

Loan Information

- Loan status
- Loan purpose
- Default indicator

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Key Analytical Measures

4	2.8%	$2,890	$27,983
Total Loan Applications	**Default Rate**	**Avg Income**	**Avg Loan Size**
Number of borrowers who applied	Percentage of loans in default	Full-time employment	Adult borrowers

Calculated measures using DAX for evaluating financial performance and risk evaluation

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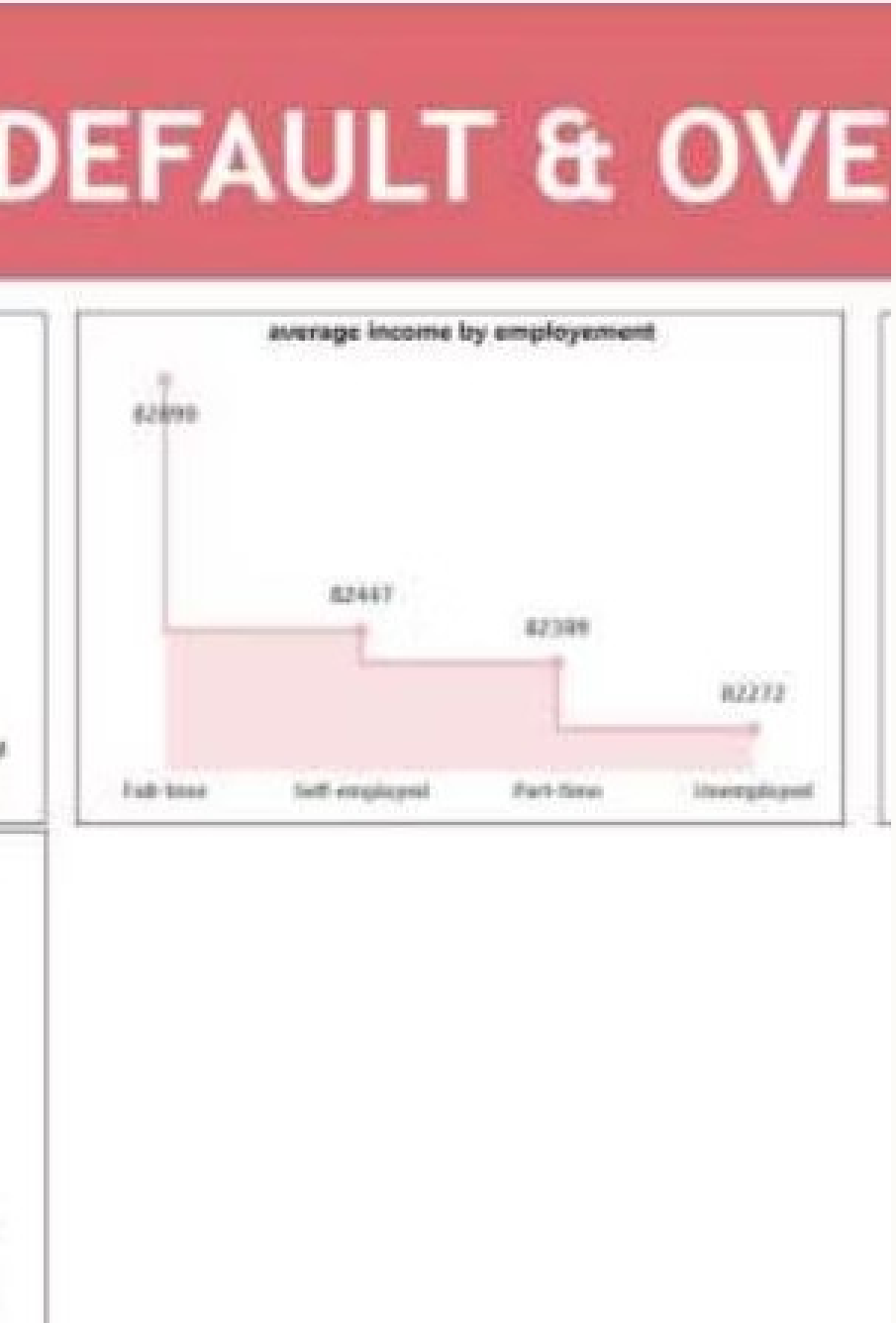

Dashboard: Default Loan Overview

Loan Amount by Purpose

House ($543M), Business ($32.2M), Education ($5.1M), Auto ($3.0M)

Average Income by Employment

Full-time ($2,890), Self-employed ($2,447), Part-time ($2,398)

Loan Amount by Age

Adults ($27,983), Middle age ($27,458), Seniors ($27,333)

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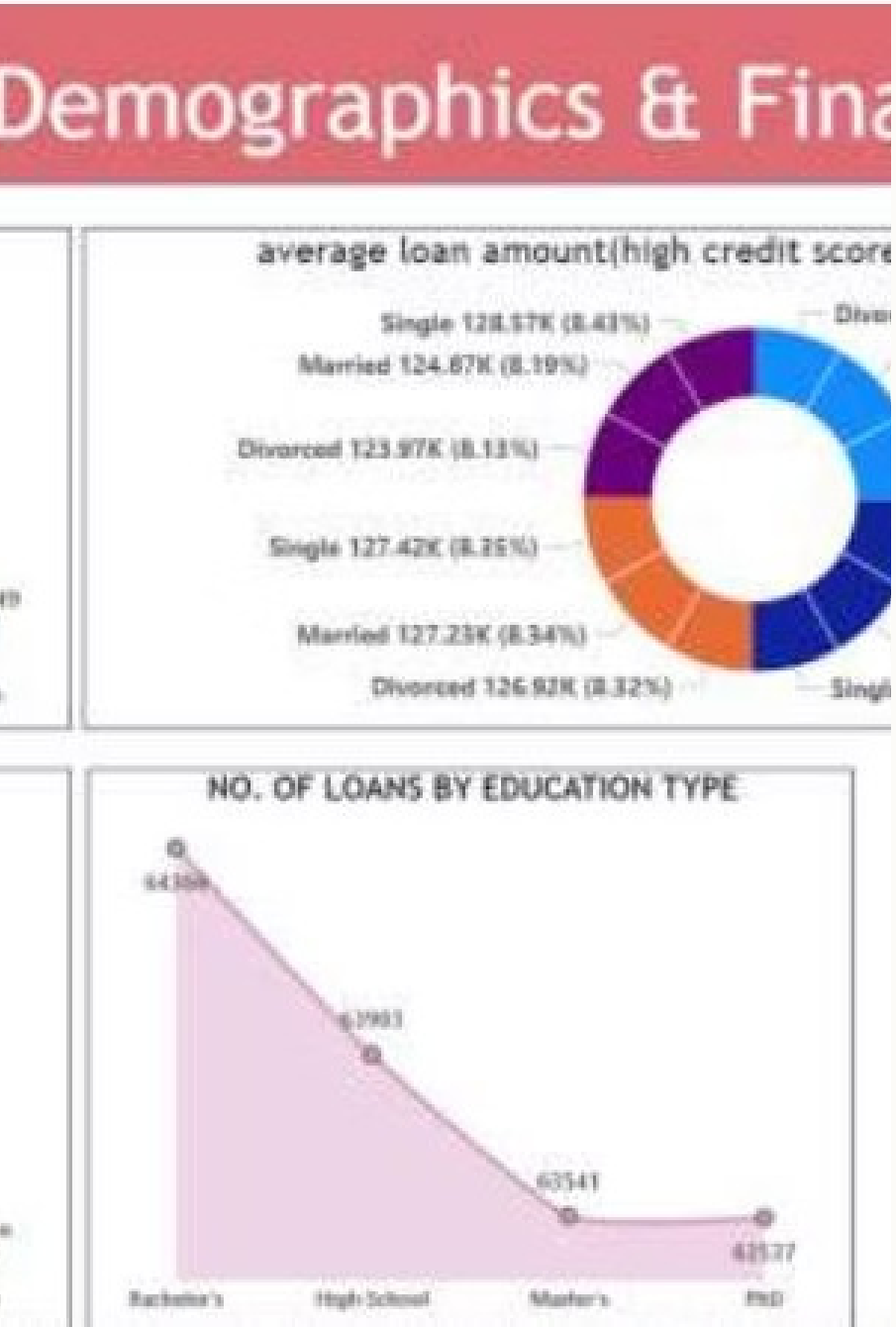

Dashboard: Demographics & Financial Profile

Key Findings

- Higher credit scores correlate with larger loan amounts
- Adults and middle-age groups show highest loan values
- Bachelor's degree holders dominate loan applications

Demographic Factors

- Age group analysis
- Employment type impact
- Education background
- Marital status patterns

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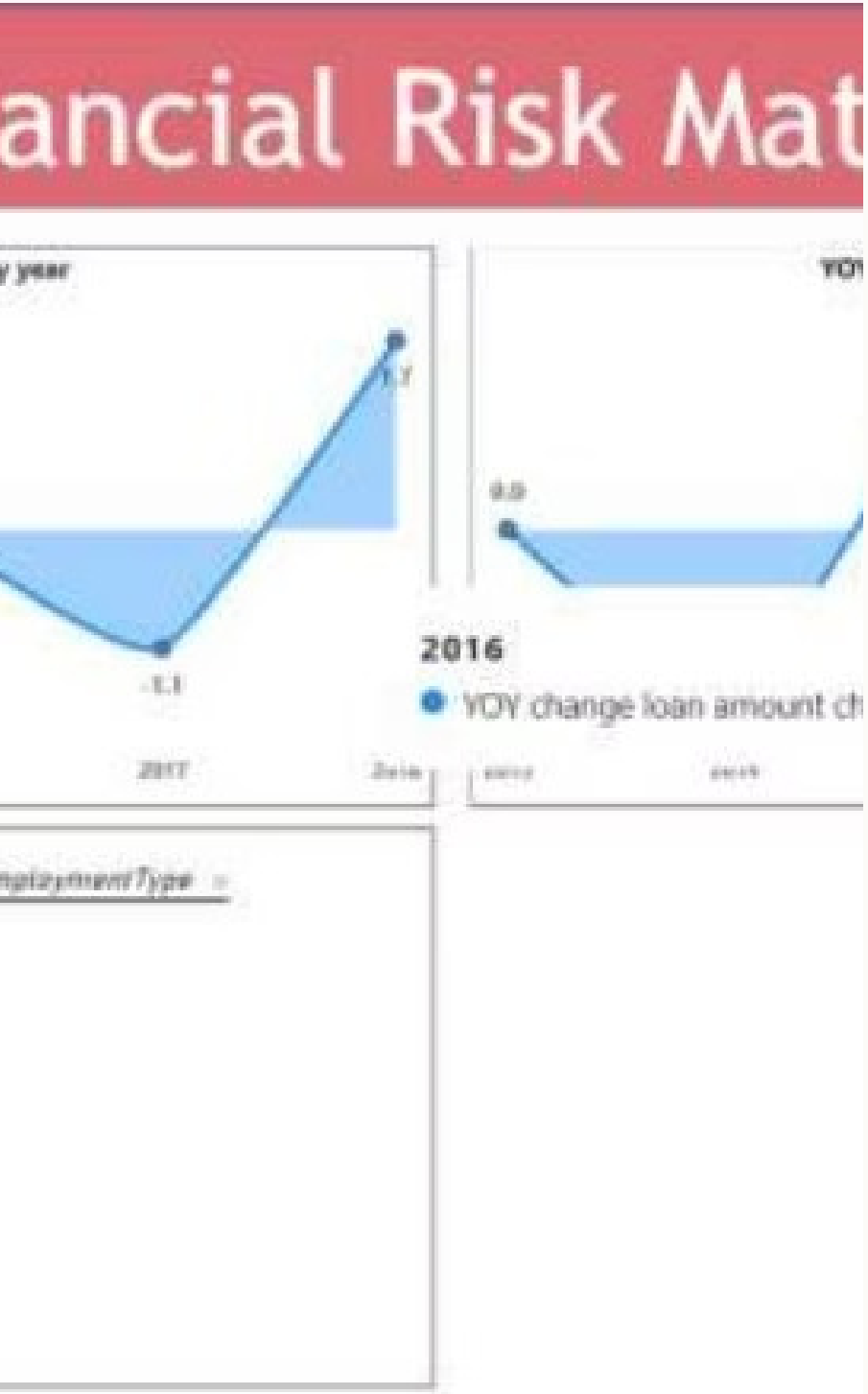

Dashboard: Financial Risk Matrix

- Creditworthiness
- Income Stability
- Loan Burden
- Repayment Capacity

YOY loan amount changes show fluctuations between -1.5% and +1.7% across years

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Key Insights

Debt-to-Income Ratio

Borrowers with higher debt relative to income demonstrate significantly higher default probability

Demographic Vulnerability

Certain demographic groups show greater financial vulnerability, requiring segmentation

Loan Size Impact

Loan amounts exceeding repayment capacity contribute heavily to default cases

Risk Segmentation

Enables prioritization of high-risk borrower monitoring and intervention

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Business Recommendations

Implement Risk-Based Loan Approval

Consider income stability and debt burden in approval process

Strengthen Credit Scoring Models

Integrate demographic and financial indicators for accuracy

Introduce Risk-Based Interest Rates

Charge higher rates to offset default risk for high-risk borrowers

Monitor High-Risk Borrowers

Develop early warning systems for financial distress signs

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